Resilient Baggage Operations
ARN · STOCKHOLM ARLANDA

TEAM GUIDE · 2026

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Hackathon 2026 · Stockholm Arlanda

Team Onboarding Guide

Visual walkthrough of the Resilient Baggage Operations system. Each section includes a live mock of what you'll see at **localhost:3000** — use this to orient yourself before running the app, or as a reference if you're not in the shared environment.

FastAPI Backend | React 18 Frontend | Couchbase Edge + Cloud | XDCR Replication | IATA Compliance | localhost:3000

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01 System Overview

What this app does and why it matters

Core idea: Airport baggage systems can't go down — even when the cloud connection fails. This app demonstrates **edge-first resilience**: a local Couchbase node keeps running independently during an outage, and automatically syncs back to the cloud the moment connectivity is restored.

ARCHITECTURE

REPLICATION

Edge Node (local)	`localhost:8091`
Main / Cloud Node	`localhost:8092`
Sync Gateway	`localhost:4984`
Backend API	`localhost:8000`
Frontend	`localhost:3000`

All bag writes go to the **Edge node first**. XDCR (Cross Datacenter Replication) continuously pushes changes to the Main/cloud node in the background.

When the connection is cut (simulated outage), the Edge keeps accepting writes locally. When restored, everything syncs automatically — zero data loss.

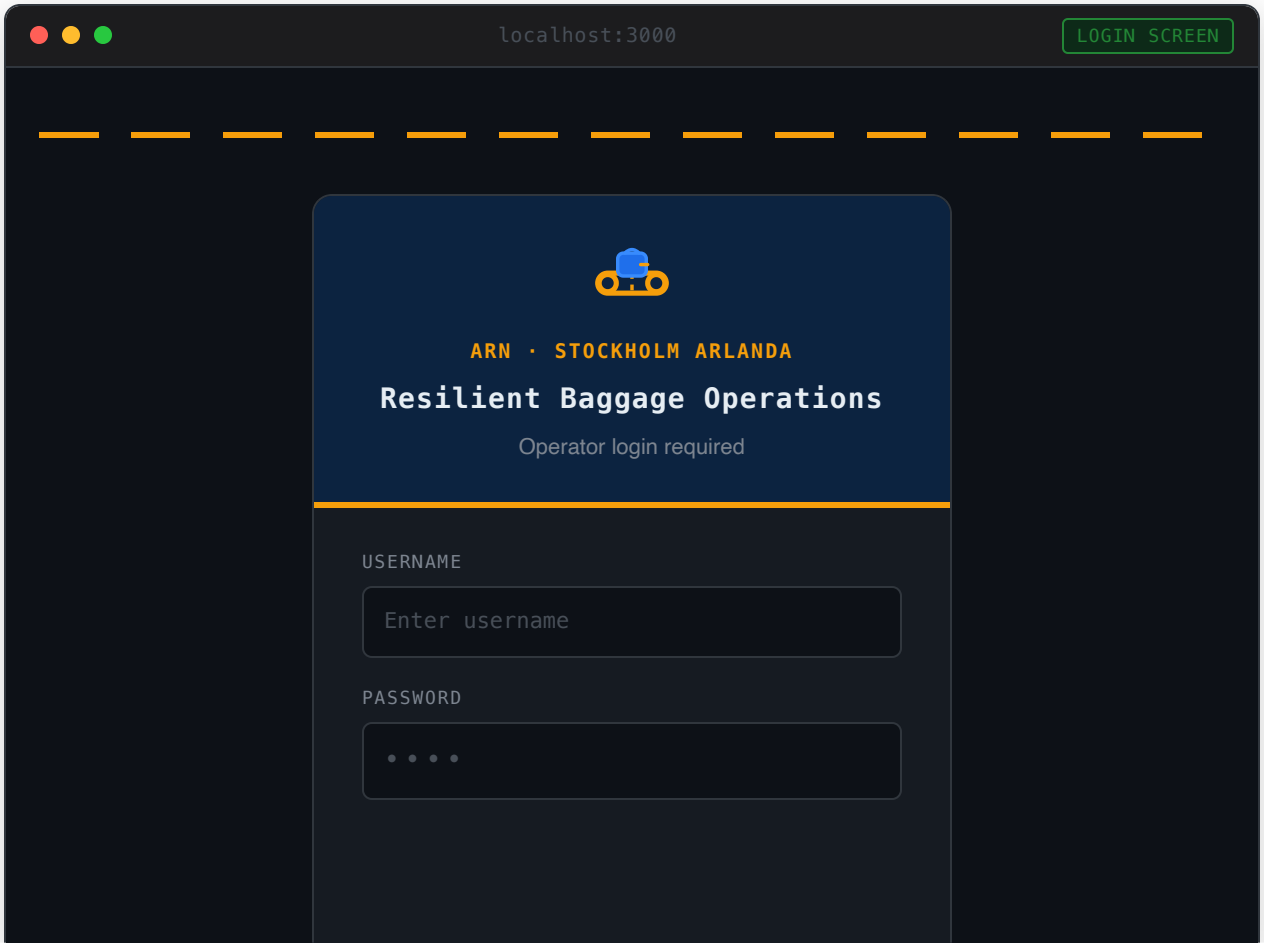
BAG STATUS LIFECYCLE

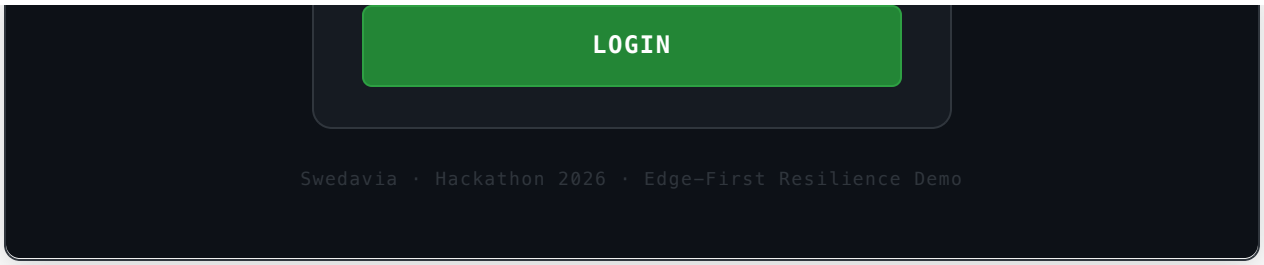
CHECK-IN → IN TRANSIT → LOADED → DELIVERED

ON HOLD ← triggered by passenger no-show (IATA rule)

02 Login Screen

How to access the operations center





CREDENTIALS

Username	aparupa · pritha · priyanka · kamali
Password	1234
Case sensitive?	No — any case works

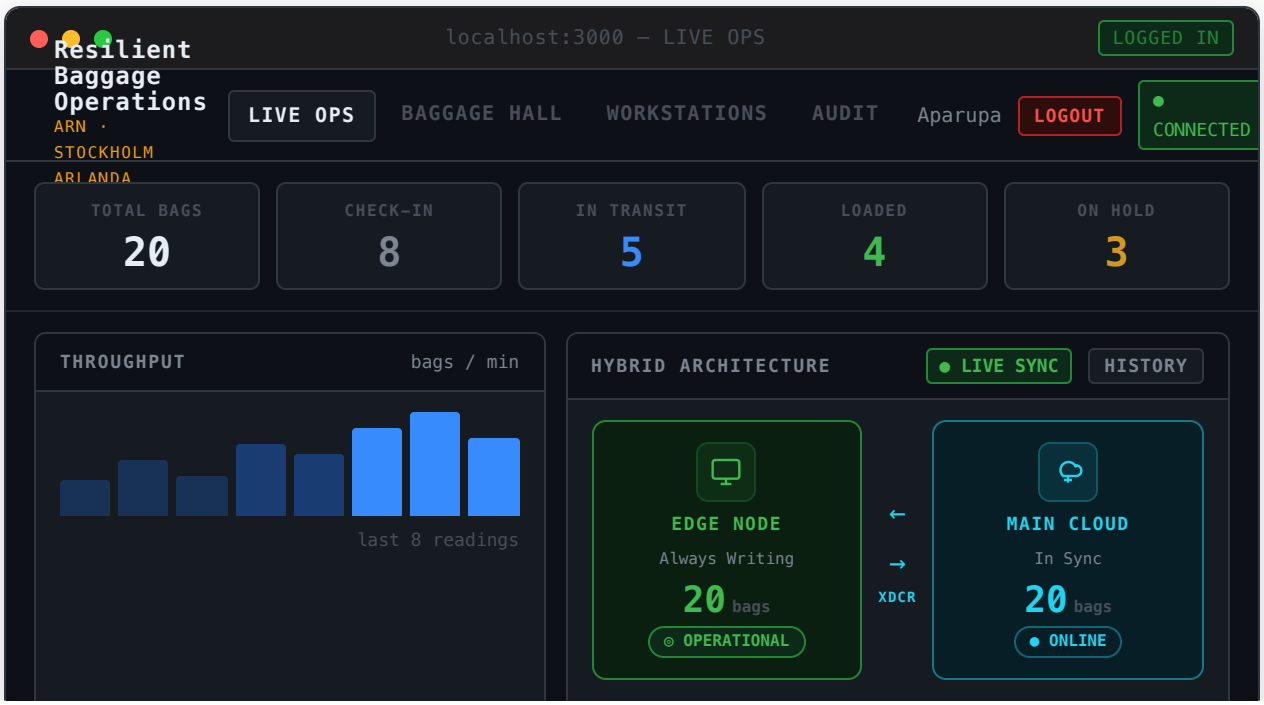
The 4 valid usernames are not shown on the login screen — only blank username + password fields. Only authorised team members know the names.

HOW THE SESSION WORKS

1. Open **localhost:3000** → login screen appears
2. Enter username + password → click **LOGIN**
3. Name saved in browser localStorage — refreshing keeps you logged in
4. To log out: click **LOGOUT** in the top-right bar
5. New browser / incognito tab = fresh login required

03 Live Ops Tab

The main operations dashboard — default landing tab after login



No active conflicts — state is consistent

All baggage data syncing to cloud in real-time

BELT HEATMAP 20 bags · 4 belts 1 JAM ▼ COLLAPSE

A1	A2	B1	B2
SK101	SK202	SK303	SK404
2 bags	4 bags	8 bags	12 bags
GREEN — Low load	YELLOW — Moderate	ORANGE — High load	RED — Near capacity
			JAM DETECTED

FLIGHT INFORMATION

Search flight / dest…

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FLIGHT	ORIG	DEST	DEP.	ARR.	GATE	BELT
SK101	ARN	CPH	06:45	—	A12	A1
SK202	ARN	LHR	08:20	—	B5	A2
SK303	ARN	FRA	09:10	—	C3	B1
SK404	ARN	AMS	05:30	07:15	D8	B2
SK505	ARN	CDG	11:55	—	E2	—

CONVEYOR SCREENING

BELT	SPEED	LOAD	TEMP	STATUS
A1	1.2 m/s	18%	22°C	OK
A2	1.1 m/s	42%	24°C	OK
B1	0.8 m/s	67%	29°C	WARN
B2	0.0 m/s	88%	41°C	JAM

KPI strip (top of tab): 5 live counters from `/bags`. **Split-Brain Panel**: Edge vs Cloud bag counts — turns red when XDCR is paused. **Belt Heatmap**: collapsible, click the header to hide/show.

04 Baggage Hall Tab

Passenger boarding management and full bag table

localhost:3000 — BAGGAGE HALL PASSENGER BOARD

PASSENGER BOARD SK101 SK202 SK303 SK404 SK505

PAX ID	NAME	FLIGHT	STATUS	ACTIONS
PAX-3BE7C1C1	Erik Johansson	SK101	CHECKED IN	BOARD NO SHOW
PAX-A1B2C3D4	Anna Lindqvist	SK101	BOARDED	— boarded —
PAX-F9E8D7C6	Magnus Eriksson	SK101	NO SHOW	— bags on hold —
PAX-22334455	Sara Persson	SK101	CHECKED IN	BOARD NO SHOW

ALL BAGS

Search passenger, bag, flight…

BAG ID	PAX ID	PASSENGER	FLIGHT	BELT	STATUS	KG	SYNC
BAG-E83C65F5	PAX-3BE7C1C1	Erik Johansson	SK101	A1	IN TRANSIT	12.4	SYNCED
BAG-C1D2E3F4	PAX-F9E8D7C6	Magnus Eriksson	SK101	—	ON HOLD	8.7	SYNCED
BAG-AABBCCDD	PAX-22334455	Sara Persson	SK202	A2	CHECK_IN	26.1 ⚖	PENDING

IATA Security Rule: Clicking NO SHOW immediately triggers a backend query that sets all of that passenger's active bags to `status=on_hold`. The AlertRail fires a HIGH severity alert automatically.

05 Workstations Tab

Checkpoint-specific operator views — 3 columns by stage

localhost:3000 — WORKSTATIONS

3 CHECKPOINTS

OPS SNAPSHOT **17 active** **3 on hold** **1 misrouted**

SYNC

WS-1 SECURITY SCREENING 8

BAG	PASSENGER	FLIGHT
BAG-AA11BB22	Anna L.	SK101
BAG-CC33DD44	Lars B.	SK202

WS-2 CONVEYOR ROUTING 5

BAG	PASSENGER	BELT
BAG-E83C65F5	Erik J.	A1
BAG-11223344	Maja H.	A2

WS-3 AIRCRAFT LOADING 4

BAG	PASSENGER	FLIGHT
BAG-99AABBCC	Sara P.	SK101
BAG-DDEEFF00	Johan K.	SK101

BAG	PASSENGER	FLIGHT
BAG-EE55FF66	Karin S.	SK303
	+5 more	

BAG	PASSENGER	BELT
BAG-55667788	Per A.	B1
	+2 more	

BAG	PASSENGER	FLIGHT
BAG-11AABB22	Elin N.	SK202
BAG-33CCDD44	Nils F.	SK202

Read-only views — no editing from here. Designed so a ground operator at each station sees only the bags at their checkpoint. Updates live every 5 seconds.

06 Audit Tab

Immutable log of all system events

localhost:3000 — AUDIT

AUDIT LOG

AUDIT LOG reverse chronological · live

Filter event type…

TIME	EVENT	ENTITY	OPERATOR	DETAIL
09:14:03	PASSENGER_NO_SHOW	PAX-F9E8D7C6	Aparupa	Magnus Eriksson — 1 bag on hold
09:12:51	PASSENGER_BOARDED	PAX-A1B2C3D4	Aparupa	Anna Lindqvist boarded SK101
09:10:22	BAG_STATUS_UPDATE	BAG-E83C65F5	System	check_in → in_transit
09:08:44	SYNC_DIVERGENCE	—	System	Edge=20 Main=17 · 3 pending
09:05:00	SEED	—	Aparupa	5 flights + 20 bags + 20 passengers

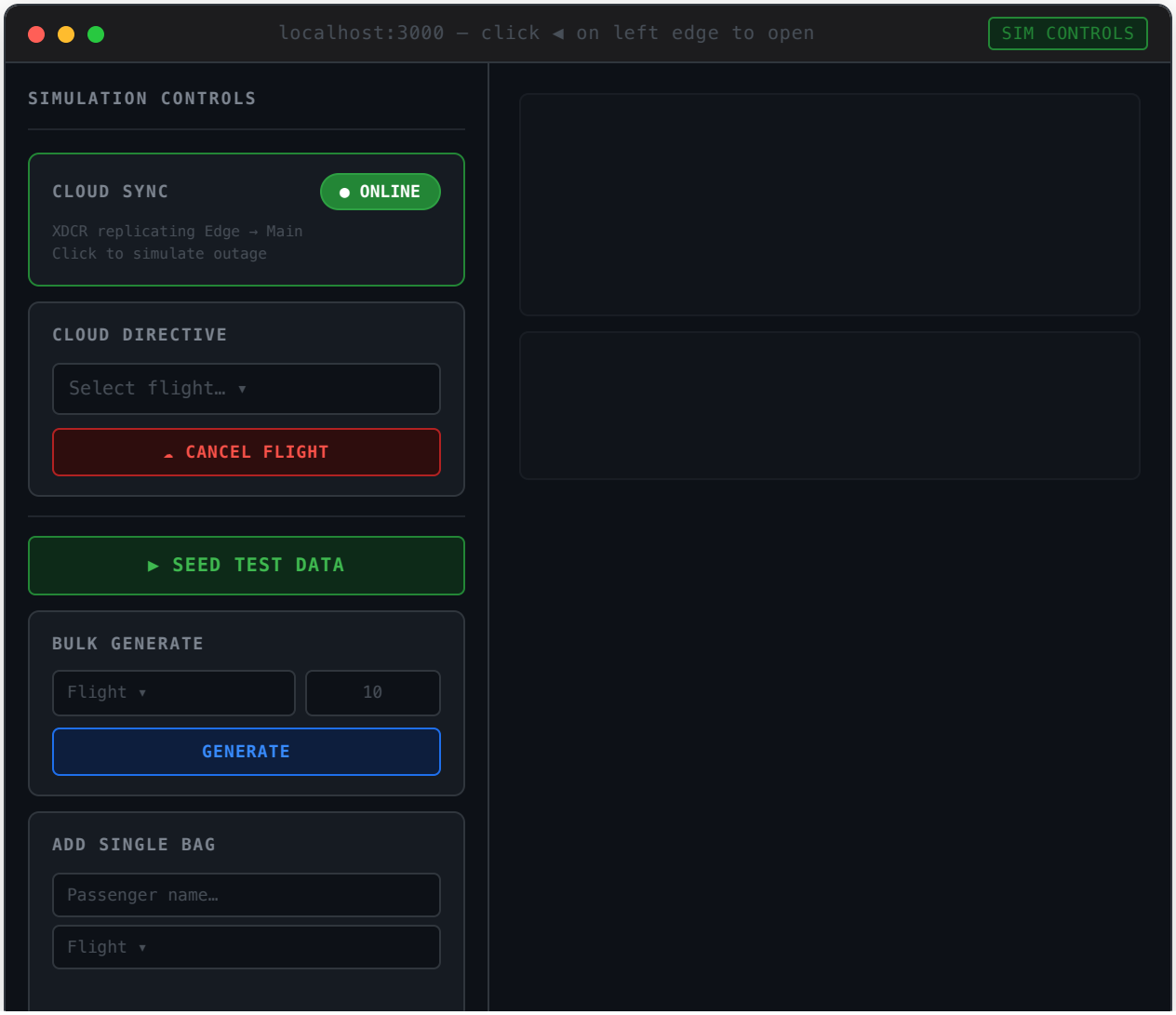
The audit log is stored directly in Couchbase and survives outages. Even writes made while the cloud is offline are logged to the Edge node and synced back once connectivity is restored.

EVENT TYPE	WHEN IT FIRES
bag_created	New bag added (manual or bulk generate)
bag_status_update	Operator changes a bag's status

EVENT TYPE	WHEN IT FIRES
`passenger_no_show`	No-show declared — bags put on hold
`passenger_boarded`	Passenger marked as boarded
`conflict_resolved`	Split-brain conflict accepted or overridden
`sync_divergence`	System detected Edge ≠ Main doc counts
`seed`	Test data seeded

07 Simulation Controls

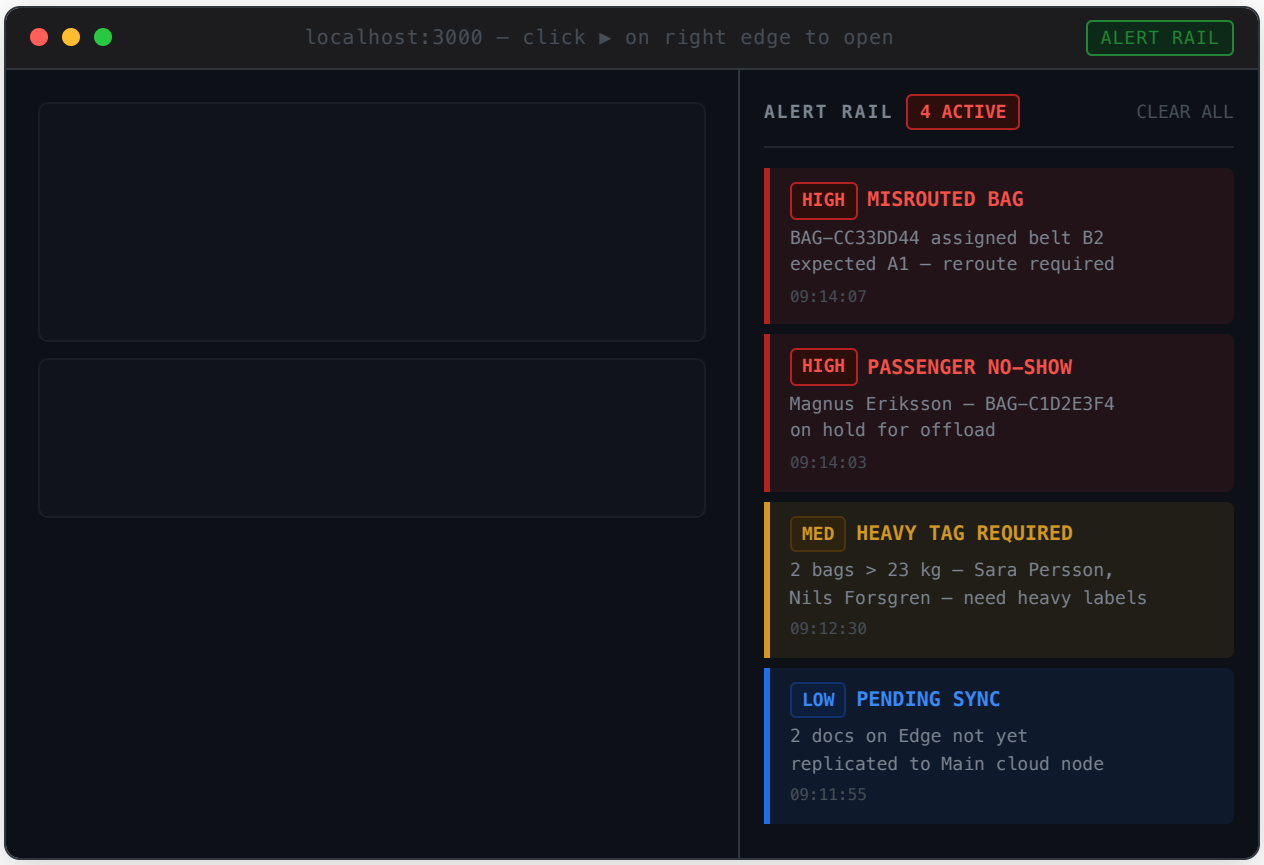
Left-side drawer — the demo engine



Click the ◀ **tab on the left edge** of the screen to open or close the drawer. The core outage demo is the **CLOUD SYNC toggle** — flip it OFFLINE to pause XDCR, then flip ONLINE to restore and watch sync happen automatically.

08 Alert Rail

Right-side drawer — live operational alerts



SEVERITY	ALERT TYPE	TRIGGER CONDITION
HIGH	MISROUTED BAG	Bag on wrong belt (misrouted flag = true)
HIGH	PASSENGER NO-SHOW	Bag on hold due to no-show rule
HIGH	CONVEYOR JAM	Belt jam detected in integrations feed
MEDIUM	HEAVY TAG REQUIRED	One or more bags > 23 kg not yet delivered
MEDIUM	NO SCAN ALERT	Bag not scanned for more than 15 minutes
MEDIUM	SYNC OFFLINE	Cloud connection is down
LOW	PENDING SYNC	Docs on Edge not yet replicated to Main

09 Key Demo Flows

The 3 most important scenarios to show a client

FLOW 1 · SEED & EXPLORE (2 MIN)

Get the system to a clean demo state and walk through the UI.

1. Open Simulation Controls (◀ left tab) → click **SEED TEST DATA**
2. Watch event log: "SEEDED 5 flights + 20 bags on Edge [SYNCING to Main]"
3. Show Split-Brain Panel: Edge = Main = 20 bags, status **IN SYNC**
4. Tour LIVE OPS → Belt Heatmap → Baggage Hall → Workstations

FLOW 2 · OUTAGE & RECOVERY (3 MIN) ← MOST IMPRESSIVE

Demonstrate that the system keeps working with zero downtime during a cloud outage.

1. Simulation Controls → toggle **CLOUD SYNC to OFFLINE**
2. Add a bag or generate load — notice `PENDING` sync label on new bags
3. Split-Brain Panel shows Edge > Main — diverged, turning red
4. Point out: **app is fully operational** — operators can still check in bags, view flights, update statuses
5. Toggle **CLOUD SYNC back to ONLINE**

6. Watch Split-Brain Panel: Edge = Main, **SYNCED** — all offline bags replicated with zero data loss

FLOW 3 · PASSENGER NO-SHOW (2 MIN)

Demonstrate IATA compliance automation.

1. Go to **BAGGAGE HALL** tab → Passenger Board
2. Pick any passenger with status **CHECKED IN**
3. Click **NO SHOW** → confirm
4. Toast appears: "N bags put on hold for offload"
5. Open AlertRail (▶ right tab) → HIGH severity: **PASSENGER NO-SHOW** alert
6. Bag Table shows affected bags with status **ON HOLD**

10 API Quick Reference

Backend endpoints at localhost:8000

METHOD	ENDPOINT	WHAT IT DOES
GET	`/health`	System health, Edge/Main doc counts, sync status
GET	`/bags`	All bags
POST	`/bags`	Create a new bag
PUT	`/bags/{id}/status`	Update bag status
GET	`/flights`	All flights
POST	`/simulate/seed`	Reset + seed 5 flights, 20 bags, 20 passengers
POST	`/simulate/offline`	Pause XDCR replication (simulate outage)
POST	`/simulate/online`	Resume XDCR replication (restore)
POST	`/simulate/load?count=N`	Bulk generate N bags

METHOD	ENDPOINT	WHAT IT DOES
GET	/passengers	All passengers (optional ?flight_id=SK101)
PUT	/passengers/{id}/board	Mark passenger as boarded
POST	/passengers/{id}/no-show	Declare no-show → auto hold all bags
GET	/audit	Audit log entries
GET	/conflicts	Unresolved split-brain conflicts
GET	/integrations/live	Conveyor belt health, DCS, ACRIS, RFID feed

To start the system:

```
cd swedavia-baggage && docker compose up -d
```

Then open **http://localhost:3000** and log in with your name + password 1234

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GitHub: **AparupaBanerjee/ResilientBaggageOperationsApp** · Docker Hub: **aparupada/resilient_baggage_operations**